

The empirical recurrence for OEIS A259999, recovered from its tail

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Abstract

OEIS A259999 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 222$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 103$. Its last coefficient is $c_{96} = 264067128 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A259999 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000011 or 00000101.”. It has offset 1 and begins

2943, 12412, 39209, 74850, 190794, 760397, 1890678, 4225837, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:58:12, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 222$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 222$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 444$ exact terms from $n = 7$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	0	c_{25}	−105420806	c_{49}	27731571026	c_{73}	−194501188157
c_2	1	c_{26}	−22237410	c_{50}	138572873864	c_{74}	3364563794744
c_3	34	c_{27}	185069741	c_{51}	531750122238	c_{75}	2522102775530
c_4	77	c_{28}	278723998	c_{52}	−432020170823	c_{76}	405830907808
c_5	−33	c_{29}	−38721172	c_{53}	−531525315686	c_{77}	−912495569166
c_6	−459	c_{30}	−1138464414	c_{54}	−74799220148	c_{78}	−1273955828997
c_7	−2395	c_{31}	319041588	c_{55}	982315721891	c_{79}	−778206291114
c_8	−1604	c_{32}	542159787	c_{56}	241491601290	c_{80}	−255367584363
c_9	4330	c_{33}	3357642500	c_{57}	−1066651909370	c_{81}	512663151429
c_{10}	26751	c_{34}	−6096522570	c_{58}	−806402041281	c_{82}	432449516853
c_{11}	53151	c_{35}	207379424	c_{59}	955944941421	c_{83}	424833026652
c_{12}	−11796	c_{36}	−4099907112	c_{60}	1445596133278	c_{84}	−61422927399
c_{13}	−175929	c_{37}	23051920157	c_{61}	−1197903145301	c_{85}	−126436856445
c_{14}	−518789	c_{38}	−16699291177	c_{62}	−2106848033142	c_{86}	−246715788708
c_{15}	−277964	c_{39}	143258134	c_{63}	196915038402	c_{87}	−53422559883
c_{16}	817855	c_{40}	−40570259269	c_{64}	3843198172429	c_{88}	52555813524
c_{17}	2651352	c_{41}	76893292989	c_{65}	1054441155175	c_{89}	45751210074
c_{18}	2503658	c_{42}	−13140997249	c_{66}	−3328320078020	c_{90}	28017935883
c_{19}	−3588975	c_{43}	25834085993	c_{67}	−4140979928966	c_{91}	−13001193279
c_{20}	−8256416	c_{44}	−165972985161	c_{68}	2464401856837	c_{92}	−484008615
c_{21}	−7572590	c_{45}	121432996364	c_{69}	5422188677723	c_{93}	−5276175408
c_{22}	21205010	c_{46}	24608571490	c_{70}	1923293812113	c_{94}	632049561
c_{23}	15777196	c_{47}	152152597284	c_{71}	−4680357144543	c_{95}	0
c_{24}	−3621878	c_{48}	−375022432775	c_{72}	−4526564426963	c_{96}	264067128

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 103$, and it is the recurrence OEIS A259999 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{95} - \dots - c_{96}$. Its constant term is $-c_{96} = -264067128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 264067128.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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